INCH-POUND

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DETAIL SPECIFICATION SHEET

# BOLT, SLEEVE, 100° TENSION, FLUSH HEAD, STRAIGHT SHANK, CRES 304 SLEEVE, TITANIUM ALLOY PIN, 95 KSI Fsu, 160 KSI Ftu

Inactive for new design after 4 April 1998.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-B-85667.

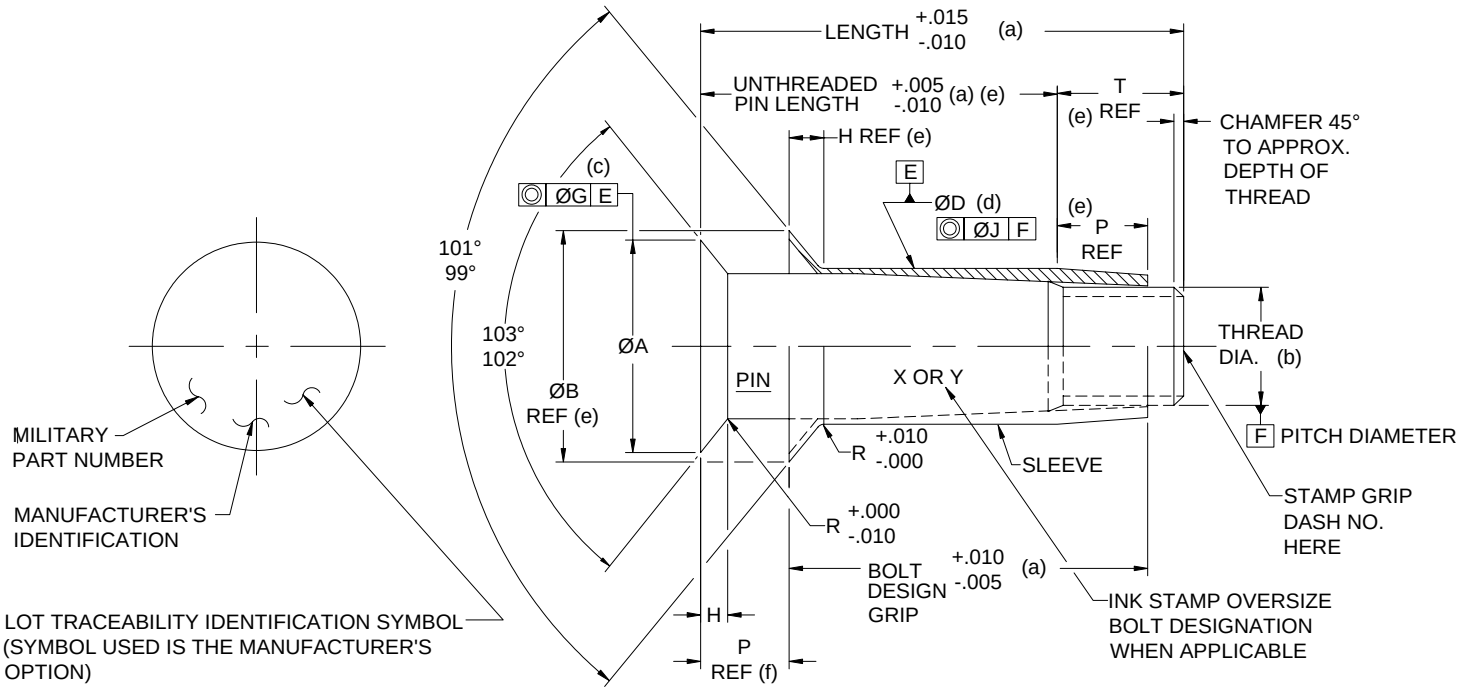



FIGURE 1. Tension flush head bolt.

(a) Grip lengths are available in .0625 increments.
(b) Threads shall be in accordance with AS8879. Thread major diameter (T.D.) shall be in accordance with Table I.
(c) 1. For sizes .190 thru .375, dimension "∅G" shall be .008 FIM.
2. For sizes .437 thru .625, .012 FIM.
3. For sizes .750, .020 FIM.
(d) 1. For sizes .190 thru .375, dimension "∅J" shall be .006 FIM.
2. For sizes .437 thru .750, .010 FIM.
(e) Reference dimensions are for design purposes only and are not an inspection or manufacturing requirement.
(f) Protrusion will vary according to taper to provide expansion as given in Table I.



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TABLE I. Bolt dimensions.

| Size Nos. | Group | Grip Dash Numbers | Thread Per SAE-AS8879 UNJF-3A | ∅A Bolt Head | ∅B Ref Sleeve Flange | | | H | F.T. +.015 (b) -.000 | P REF | R Fillet Radius | T. D. Thread Dia. | T Thread Length | ∅D Dimension Uninstalled Sleeve (a) | | | ∅D Dimension Expanded Sleeve (a)(b) | | | Minimum Installation Hole Size Diameter | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | Std. | X O/S | Y O/S | | | | | | | Std. | X O/S | Y O/S | Std. | X O/S | Y O/S | Std. | X O/S | Y O/S |
| -3 | 1 | -7 thru –10 | .1900-32 | .418<br>.369 | .457 | .463 | .469 | .083<br>.081 | .093 | .225 | .030 | .1840<br>.1810 | .343 | .2365<br>.2360 | .2425<br>.2420 | .2485<br>.2480 | .2455<br>.2445 | .2515<br>.2505 | .2575<br>.2565 | .237 | .243 | .249 |
| | 2 | -11 thru -16 | | .420<br>.370 | .459 | .465 | .471 | | .093 | .432 | | | | .2365<br>.2360 | .2425<br>.2420 | .2485<br>.2480 | .2455<br>.2445 | .2515<br>.2505 | .2575<br>.2565 | .237 | .243 | .249 |
| | 3 | -17 thru -22 | | .430<br>.380 | .476 | .483 | .490 | | .095 | .432 | | | | .2485<br>.2480 | .2555<br>.2550 | .2625<br>.2620 | .2575<br>.2565 | .2645<br>.2635 | .2715<br>.2705 | .249 | .256 | .263 |
| -4 | 1 | -8 thru -10 | .2500-28 | .550<br>.494 | .586 | .597 | .603 | .110<br>.108 | .120 | .225 | .030 | .2440<br>.2410 | .406 | .3005<br>.3000 | .3115<br>.3110 | .3175<br>.3170 | .3095<br>.3085 | .3205<br>.3195 | .3265<br>.3255 | .301 | .312 | .318 |
| | 2 | -11 thru -17 | | .553<br>.496 | .588 | .599 | .605 | | .120 | .432 | | | | .3005<br>.3000 | .3115<br>.3110 | .3175<br>.3170 | .3095<br>.3085 | .3205<br>.3195 | .3265<br>.3255 | .301 | .312 | .318 |
| | 3 | -18 thru -28 | | .566<br>.515 | .607 | .614 | .623 | | .123 | .432 | | | | .3145<br>.3140 | .3215<br>.3210 | .3305<br>.3300 | .3235<br>.3225 | .3305<br>.3295 | .3395<br>.3385 | .315 | .322 | .331 |
| -5 | 1 | -8 thru -11 | .3125-24 | .684<br>.618 | .717 | .724 | .731 | .138<br>.136 | .150 | .225 | .040 | .3060<br>.3020 | .484 | .3665<br>.3660 | .3735<br>.3730 | .3805<br>.3800 | .3755<br>.3745 | .3825<br>.3815 | .3895<br>.3885 | .367 | .374 | .381 |
| | 2 | -12 thru -20 | | .688<br>.622 | .719 | .726 | .733 | | .148 | .432 | | | | .3665<br>.3660 | .3735<br>.3730 | .3805<br>.3800 | .3755<br>.3745 | .3825<br>.3815 | .3895<br>.3885 | .367 | .374 | .381 |
| | 3 | -21 thru -31 | | .702<br>.636 | .740 | .748 | .755 | | .151 | .432 | | | | .3805<br>.3800 | .3885<br>.3880 | .3955<br>.3950 | .3895<br>.3885 | .3975<br>.3965 | .4045<br>.4035 | .381 | .389 | .396 |
| -6 | 1 | -9 thru -11 | .3750-24 | .819<br>.746 | .848 | .856 | .868 | .166<br>.164 | .175 | .225 | .040 | .3680<br>.3640 | .562 | .4315<br>.4310 | .4395<br>.4390 | .4515<br>.4510 | .4405<br>.4395 | .4485<br>.4475 | .4605<br>.4595 | .432 | .440 | .452 |
| | 2 | -12 thru -22 | | .823<br>.750 | .851 | .859 | .871 | | .176 | .432 | | | | .4315<br>.4310 | .4395<br>.4390 | .4515<br>.4510 | .4405<br>.4395 | .4485<br>.4475 | .4605<br>.4595 | .432 | .440 | .452 |
| | 3 | -23 thru -34 | | .838<br>.765 | .872 | .880 | .888 | | .178 | .432 | | | | .4475<br>.4470 | .4555<br>.4550 | .4635<br>.4630 | .4565<br>.4555 | .4645<br>.4635 | .4725<br>.4715 | .448 | .456 | .464 |
| -7 | 1 | -9 thru -12 | .4375-20 | .956<br>.875 | .987 | .999 | 1.007 | .194<br>.192 | .205 | .225 | .050 | .4310<br>.4260 | .625 | .4985<br>.4980 | .5105<br>.5100 | .5185<br>.5180 | .5075<br>.5065 | .5195<br>.5185 | .5275<br>.5265 | .499 | .511 | .519 |
| | 2 | -13 thru -26 | | .960<br>.878 | .991 | 1.003 | 1.011 | | .207 | .432 | | | | .4985<br>.4980 | .5105<br>.5100 | .5185<br>.5180 | .5075<br>.5065 | .5195<br>.5185 | .5275<br>.5265 | .499 | .511 | .519 |
| | 3 | -27 thru -40 | | .978<br>.895 | 1.015 | 1.026 | 1.038 | | .208 | .432 | | | | .5185<br>.5180 | .5295<br>.5290 | .5415<br>.5410 | .5275<br>.5265 | .5385<br>.5375 | .5505<br>.5495 | .519 | .530 | .542 |
| -8 | 1 | -10 thru -12 | .5000-20 | 1.093<br>1.005 | 1.120 | 1.130 | 1.140 | .222<br>.220 | .231 | .225 | .050 | .4930<br>.4880 | .703 | .5695<br>.5690 | .5795<br>.5790 | .5895<br>.5890 | .5785<br>.5775 | .5885<br>.5875 | .5985<br>.5975 | .570 | .580 | .590 |
| | 2 | -13 thru -30 | | 1.097<br>1.008 | 1.125 | 1.135 | 1.145 | | .233 | .432 | | | | .5695<br>.5690 | .5795<br>.5790 | .5895<br>.5890 | .5785<br>.5775 | .5885<br>.5875 | .5985<br>.5975 | .570 | .580 | .590 |
| | 3 | -31 thru -46 | | 1.118<br>1.035 | 1.152 | 1.162 | 1.171 | | .236 | .432 | | | | .5895<br>.5890 | .5995<br>.5990 | .6085<br>.6080 | .5985<br>.5975 | .6085<br>.6075 | .6175<br>.6165 | .590 | .600 | .609 |
| -9 | 1 | -14 thru -23 | .5625-18 | 1.195<br>1.100 | 1.232 | 1.247 | 1.260 | .240<br>.238 | .249 | .432 | .070 | .5550<br>.5500 | .781 | .6395<br>.6390 | .6545<br>.6540 | .6675<br>.6670 | .6485<br>.6475 | .6635<br>.6625 | .6765<br>.6755 | .640 | .655 | .668 |
| | 2 | -24 thru -48 | | 1.226<br>1.131 | 1.274 | 1.290 | 1.305 | | .253 | .432 | | | | .6705<br>.6700 | .6865<br>.6860 | .7015<br>.7010 | .6795<br>.6785 | .6955<br>.6945 | .7105<br>.7095 | .671 | .687 | .702 |
| | 3 | -49 thru -72 | | 1.226<br>1.131 | 1.274 | 1.290 | 1.305 | | .253 | .900 | | | | .6705<br>.6700 | .6865<br>.6860 | .7015<br>.7010 | .6795<br>.6785 | .6955<br>.6945 | .7105<br>.7095 | .671 | .687 | .702 |
| -10 | 1 | -14 thru -25 | .6250-18 | 1.298<br>1.195 | 1.355 | 1.374 | 1.394 | .255<br>.253 | .272 | .432 | .073 | .6160<br>.6120 | .843 | .7075<br>.7070 | .7265<br>.7260 | .7465<br>.7460 | .7165<br>.7155 | .7355<br>.7345 | .7555<br>.7545 | .708 | .727 | .747 |
| | 2 | -26 thru -52 | | 1.332<br>1.231 | 1.389 | 1.407 | 1.422 | | .270 | .432 | | | | .7465<br>.7460 | .7645<br>.7640 | .7795<br>.7790 | .7555<br>.7545 | .7735<br>.7725 | .7885<br>.7875 | .747 | .765 | .780 |
| | 3 | -53 thru -78 | | 1.332<br>1.231 | 1.389 | 1.407 | 1.422 | | .270 | .900 | | | | .7465<br>.7460 | .7645<br>.7640 | .7795<br>.7790 | .7555<br>.7545 | .7735<br>.7725 | .7885<br>.7875 | .747 | .765 | .780 |
| -12 | 1 | -14 thru -29 | .7500-16 | 1.492<br>1.389 | 1.549 | 1.564 | 1.580 | .280<br>.278 | .296 | .432 | .073 | .7430<br>.7370 | 1.000 | .8425<br>.8420 | .8575<br>.8570 | .8735<br>.8730 | .8515<br>.8505 | .8665<br>.8655 | .8825<br>.8815 | .843 | .858 | .874 |
| | 2 | -30 thru -60 | | 1.531<br>1.417 | 1.588 | 1.608 | 1.624 | | .295 | .432 | | | | .8845<br>.8840 | .9045<br>.9040 | .9205<br>.9200 | .8935<br>.8925 | .9135<br>.9125 | .9295<br>.9285 | .885 | .905 | .921 |
| | 3 | -61 thru -90 | | 1.531<br>1.417 | 1.588 | 1.608 | 1.624 | | .295 | .900 | | | | .8845<br>.8840 | .9045<br>.9040 | .9205<br>.9200 | .8935<br>.8925 | .9135<br>.9125 | .9295<br>.9285 | .885 | .905 | .921 |

(a) ∅D shall be verified within the grip bearing portion engaging the bolt shank prior to expansion; when bolt is installed to sleeve flange the diameter shall be within the expanded ∅D. Limits shown in Table I.

(b) The sleeve bolt is manufactured as a precision assembly and the components should never be dismantled. In the event of inadvertent separation, please replace with a new assembly



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TABLE II. Mechanical properties.

| Dia Dash Number | Ultimate Tensile Strength Lbs-Min (b) | Thread Area $In^2$ (b) | Double Shear Strength Lbs-Min (a) | Pin Test Diameter (± .0005) In (a) | Pin Test Dia Area $In^2$ (a) | Max Install Torque In-Lbs | Fatigue Test Loading Lbs ± 2% (c) | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | Max | Min |
| -3 | 3,620 | .0226 | 5,380 | .1900 | .0284 | 60 | 1,450 | 145 |
| -4 | 6,470 | .0404 | 9,300 | .2500 | .0491 | 115 | 2,590 | 259 |
| -5 | 10,200 | .0640 | 14,600 | .3125 | .0767 | 185 | 4,080 | 408 |
| -6 | 15,200 | .0951 | 21,000 | .3750 | .1104 | 335 | 6,080 | 608 |
| -7 | 20,600 | .1288 | 28,600 | .4375 | .1503 | 540 | 8,240 | 824 |
| -8 | 27,500 | .1717 | 37,300 | .5000 | .1963 | 805 | 11,000 | 1,100 |
| -9 | 34,800 | .2176 | 47,200 | .5625 | .2485 | 1080 | 13,920 | 1,392 |
| -10 | 43,600 | .2724 | 58,300 | .6250 | .3068 | 1560 | 17,440 | 1,744 |
| -12 | 63,200 | .3953 | 83,900 | .7500 | .4418 | 2930 | 25,280 | 2,528 |

(a) Shear test performed on pins ground to pin test diameter. The shear values are based on a minimum shear strength of 95 ksi and the stress area at the test diameter.
(b) The ultimate tensile strength values are calculated on the basic pitch diameter of the external thread and a tensile strength of 160 ksi.
(c) Maximum fatigue load is based on 40% of the ultimate tensile strength. Minimum fatigue load is 10% of maximum fatigue load.

TABLE III. Bolt grips and lengths.

| Grip Number | Bolt Design Grip | -3 Diameter | | -4 Diameter | | -5 Diameter | |
|---|---|---|---|---|---|---|---|
| | | Unthreaded Pin Length (See Figure 1) | Bolt Length | Unthreaded Pin Length (See Figure 1) | Bolt Length | Unthreaded Pin Length (See Figure 1) | Bolt Length |
| 6 | .375 | .405 | .748 | .405 | .811 | .405 | .889 |
| 7 | .438 | .468 | .811 | .468 | .874 | .468 | .952 |
| 8 | .500 | .530 | .873 | .530 | .936 | .530 | 1.014 |
| 9 | .562 | .592 | .935 | .592 | .998 | .592 | 1.076 |
| 10 | .625 | .655 | .998 | .655 | 1.061 | .655 | 1.139 |
| 11 | .688 | .718 | 1.061 | .718 | 1.124 | .718 | 1.202 |
| 12 | .750 | .780 | 1.123 | .780 | 1.186 | .780 | 1.264 |
| 13 | .812 | .842 | 1.185 | .842 | 1.248 | .842 | 1.326 |
| 14 | .875 | .905 | 1.248 | .905 | 1.311 | .905 | 1.389 |
| 15 | .938 | .968 | 1.311 | .968 | 1.374 | .968 | 1.452 |
| 16 | 1.000 | 1.030 | 1.373 | 1.030 | 1.436 | 1.030 | 1.514 |
| 17 | 1.062 | 1.092 | 1.435 | 1.092 | 1.498 | 1.092 | 1.576 |
| 18 | 1.125 | 1.155 | 1.498 | 1.155 | .1.561 | 1.155 | 1.639 |
| 19 | 1.188 | 1.218 | 1.561 | 1.218 | 1.624 | 1.218 | 1.702 |
| 20 | 1.250 | 1.280 | 1.623 | 1.280 | 1.686 | 1.280 | 1.764 |
| 21 | 1.312 | 1.234 | 1.685 | 1.234 | 1.748 | 1.342 | 1.826 |
| 22 | 1.375 | 1.405 | 1.748 | 1.405 | 1.811 | 1.405 | 1.889 |
| 23 | 1.438 | 1.468 | 1.811 | 1.468 | 1.874 | 1.468 | 1.952 |
| 24 | 1.500 | 1.530 | 1.873 | 1.530 | 1.936 | 1.530 | 2.014 |
| 25 | 1.562 | | | 1.592 | 1.998 | 1.592 | 2.076 |
| 26 | 1.625 | | | 1.655 | 2.061 | 1.655 | 2.139 |
| 27 | 1.688 | | | 1.718 | 2.124 | 1.718 | 2.702 |
| 28 | 1.750 | | | 1.780 | 2.186 | 1.780 | 2.264 |
| 29 | 1.812 | | | | | 1.842 | 2.326 |
| 30 | 1.875 | | | | | 1.905 | 2.389 |
| 31 | 1.938 | | | | | 1.968 | 2.452 |



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TABLE III. Bolt grips and lengths (continued).

| Grip Number | Bolt Design Grip | -6 Diameter | | -7 Diameter | | -8 Diameter | |
|---|---|---|---|---|---|---|---|
| | | Unthreaded Pin Length (See Figure 1) | Bolt Length | Unthreaded Pin Length (See Figure 1) | Bolt Length | Unthreaded Pin Length (See Figure 1) | Bolt Length |
| 6 | .375 | .405 | .968 | .405 | 1.030 | .405 | 1.108 |
| 7 | .438 | .468 | 1.030 | .468 | 1.093 | .468 | 1.171 |
| 8 | .500 | .530 | 1.092 | .530 | 1.155 | .530 | 1.233 |
| 9 | .562 | .592 | 1.154 | .592 | 1.217 | .592 | 1.295 |
| 10 | .625 | .655 | 1.217 | .655 | 1.280 | .655 | 1.358 |
| 11 | .688 | .718 | 1.280 | .718 | 1 .343 | .718 | 1.421 |
| 12 | .750 | .780 | 1.342 | .780 | 1.405 | .780 | 1.483 |
| 13 | .812 | .842 | 1.404 | .842 | 1.467 | .842 | 1.545 |
| 14 | .875 | .905 | 1.467 | .905 | 1.530 | .905 | 1.608 |
| 15 | .938 | .968 | 1.530 | .968 | 1.593 | .968 | 1.671 |
| 16 | 1.000 | 1.030 | 1.592 | 1.030 | 1.655 | 1.030 | 1.733 |
| 17 | 1.062 | 1.092 | 1.654 | 1.092 | 1.717 | 1.092 | 1.795 |
| 18 | 1.125 | 1.155 | 1.717 | 1.155 | 1.780 | 1.155 | 1.858 |
| 19 | 1.188 | 1.218 | 1.780 | 1.218 | 1.843 | 1.218 | 1.921 |
| 20 | 1.250 | 1.280 | 1.842 | 1.280 | 1.905 | 1.280 | 1.983 |
| 21 | 1.312 | 1.342 | 1.904 | 1.342 | 1.967 | 1.342 | 2.045 |
| 22 | 1.375 | 1.405 | 1.967 | 1.405 | 2.030 | 1.405 | 2.108 |
| 23 | 1.438 | 1.468 | 2.030 | 1.468 | 2.093 | 1.468 | 2.171 |
| 24 | 1.500 | 1.530 | 2.092 | 1.530 | 2.155 | 1.530 | 2.233 |
| 25 | 1.562 | 1.592 | 2.154 | 1.592 | 2.217 | 1.592 | 2.295 |
| 26 | 1.625 | 1.655 | 2.217 | 1.655 | 2.280 | 1.655 | 2.358 |
| 27 | 1.688 | 1.718 | 2.280 | 1.718 | 2.343 | 1.718 | 2.421 |
| 28 | 1.750 | 1.780 | 2.342 | 1.780 | 2.405 | 1.780 | 2.483 |
| 29 | 1.812 | 1.842 | 2.404 | 1.842 | 2.467 | 1.842 | 2.545 |
| 30 | 1.875 | 1.905 | 2.467 | 1.905 | 2.530 | 1.905 | 2.608 |
| 31 | 1.938 | 1.968 | 2.530 | 1.968 | 2.593 | 1.968 | 2.671 |
| 32 | 2.000 | 2.030 | 2.592 | 2.030 | 2.655 | 2.030 | 2.733 |
| 33 | 2.062 | 2.092 | 2.654 | 2.092 | 2.717 | 2.092 | 2.795 |
| 34 | 2.125 | 2.155 | 2.717 | 2.155 | 2.780 | 2.155 | 2.858 |
| 35 | 2.188 | | | 2.218 | 2.843 | 2.218 | 2.921 |
| 36 | 2.250 | | | 2.280 | 2.905 | 2.280 | 2.983 |
| 37 | 2.312 | | | 2.342 | 2.967 | 2.342 | 3.045 |
| 38 | 2.375 | | | 2.405 | 3.030 | 2.405 | 3.108 |
| 39 | 2.438 | | | 2.468 | 3.093 | 2.468 | 3.171 |
| 40 | 2.500 | | | 2.530 | 3.155 | 2.530 | 3.233 |
| 41 | 2.562 | | | | | 2.592 | 3.295 |
| 42 | 2.625 | | | | | 2.655 | 3.358 |
| 43 | 2.688 | | | | | 2.718 | 3.421 |
| 44 | 2.750 | | | | | 2.780 | 3.483 |
| 45 | 2.812 | | | | | 2.842 | 3.545 |
| 46 | 2.875 | | | | | 2.905 | 3.608 |



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TABLE III. Bolt grips and lengths (continued).

| Grip Number | -9 Diameter | | -10 Diameter | | -12 Diameter | |
|---|---|---|---|---|---|---|
| | Unthreaded Pin Length (See Figure 1) | Bolt Length | Unthreaded Pin Length (See Figure 1) | Bolt Length | Unthreaded Pin Length (See Figure 1) | Bolt Length |
| 14 | .905 | 1.686 | .905 | 1.748 | .905 | 1.905 |
| 15 | .968 | 1.748 | .968 | 1.811 | .968 | 1.968 |
| 16 | 1.030 | 1.811 | 1.030 | 1.873 | 1.030 | 2.030 |
| 17 | 1.092 | 1.874 | 1.092 | 1.935 | 1.092 | 2.092 |
| 18 | 1.155 | 1.936 | 1.155 | 1.998 | 1.155 | 2.155 |
| 19 | 1.218 | 1.998 | 1.218 | 2.061 | 1.218 | 2.218 |
| 20 | 1.280 | 2.061 | 1.280 | 2.123 | 1.280 | 2.280 |
| 21 | 1.342 | 2.124 | 1.342 | 2.185 | 1.342 | 2.342 |
| 22 | 1.405 | 2.186 | 1.405 | 2.248 | 1.405 | 2.405 |
| 23 | 1.468 | 2.248 | 1.468 | 2.311 | 1.468 | 2.468 |
| 24 | 1.530 | 2.311 | 1.530 | 2.373 | 1.530 | 2.530 |
| 25 | 1.592 | 2.374 | 1.592 | 2.435 | 1.592 | 2.592 |
| 26 | 1.655 | 2.436 | 1.655 | 2.498 | 1.655 | 2.655 |
| 27 | 1.718 | 2.498 | 1.718 | 2.561 | 1.718 | 2.718 |
| 28 | 1.780 | 2,561 | 1.780 | 2.623 | 1.780 | 2.780 |
| 29 | 1.842 | 2.624 | 1.842 | 2.685 | 1.842 | 2.842 |
| 30 | 1.905 | 2.686 | 1.905 | 2.748 | 1.905 | 2.905 |
| 31 | 1.968 | 2.748 | 1.968 | 2.811 | 1.968 | 2.968 |
| 32 | 2.030 | 2.811 | 2.030 | 2.873 | 2.030 | 3.030 |
| 33 | 2.092 | 2.874 | 2.092 | 2.935 | 2.092 | 3.092 |
| 34 | 2.155 | 2.936 | 2.155 | 2.998 | 2.155 | 3.155 |
| 35 | 2.218 | 2.998 | 2.218 | 3.061 | 2.218 | 3.218 |
| 36 | 2.280 | 3.061 | 2.280 | 3.123 | 2.280 | 3.280 |
| 37 | 2.342 | 3.124 | 2.342 | 3.185 | 2.342 | 3.342 |
| 38 | 2.405 | 3.186 | 2.405 | 3.248 | 2.405 | 3.405 |
| 39 | 2.468 | 3.248 | 2.468 | 3.311 | 2.468 | 3.468 |
| 40 | 2.530 | 3.311 | 2.530 | 3.373 | 2.530 | 3.530 |
| 41 | 2.592 | 3.374 | 2.592 | 3.435 | 2.592 | 3.592 |
| 42 | 2.655 | 3.436 | 2.655 | 3.498 | 2.655 | 3.655 |
| 43 | 2.718 | 3.498 | 2.718 | 3.561 | 2.718 | 3.718 |
| 44 | 2.780 | 3.561 | 2.780 | 3.623 | 2.780 | 3.780 |
| 45 | 2.842 | 3.624 | 2.842 | 3.685 | 2.842 | 3.842 |
| 46 | 2.905 | 3.686 | 2.905 | 3.748 | 2.905 | 3.905 |
| 47 | 2.968 | 3.748 | 2.968 | 3.811 | 2.968 | 3.968 |
| 48 | 3.030 | 3.811 | 3.030 | 3.873 | 3.030 | 4.030 |
| 49 | 3.092 | 3.874 | 3.092 | 3.935 | 3.092 | 4.092 |
| 50 | 3.155 | 3.896 | 3.155 | 3.998 | 3.155 | 4.155 |
| 51 | 3.218 | 3.998 | 3.218 | 4.061 | 3.218 | 4.218 |
| 52 | 3.280 | 4.061 | 3.280 | 4.123 | 3.280 | 4.280 |



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TABLE III. Bolt grips and lengths (continued).

| Grip Number | -9 Diameter | | -10 Diameter | | -12 Diameter | |
|---|---|---|---|---|---|---|
| | Unthreaded Pin Length (See Figure 1) | Bolt Length | Unthreaded Pin Length (See Figure 1) | Bolt Length | Unthreaded Pin Length (See Figure 1) | Bolt Length |
| 53 | 3.342 | 4.124 | 3.342 | 4.185 | 3.342 | 4.342 |
| 54 | 3.405 | 4.186 | 3.405 | 4.248 | 3.405 | 4.405 |
| 55 | 3.468 | 4.248 | 3.468 | 4.311 | 3.468 | 4.468 |
| 56 | 3.530 | 4.311 | 3.630 | 4.373 | 3.530 | 4.350 |
| 57 | 3.592 | 4.374 | 3.592 | 4.435 | 3.592 | 4.592 |
| 58 | 3.655 | 4.436 | 3.655 | 4.498 | 3.655 | 4.655 |
| 59 | 3.718 | 4.499 | 3.718 | 4.561 | 3.718 | 4.718 |
| 60 | 3.780 | 4.561 | 3.780 | 4.623 | 3.780 | 4.780 |
| 61 | 3.843 | 4.624 | 3.843 | 4.889 | 3.843 | 4.843 |
| 62 | 3.905 | 4.686 | 3.905 | 4.748 | 3.905 | 4.905 |
| 63 | 3.968 | 4.749 | 3.968 | 4.811 | 3.968 | 4.968 |
| 64 | 4.030 | 4.811 | 4.030 | 4.873 | 4.030 | 5.030 |
| 65 | 4.093 | 4.874 | 4.093 | 4.936 | 4.093 | 5.093 |
| 66 | 4.155 | 4.936 | 4.155 | 4.998 | 4.155 | 5.155 |
| 67 | 4.218 | 4.999 | 4.218 | 5.061 | 4.218 | 5.218 |
| 68 | 4.280 | 5.061 | 4.280 | 5.123 | 4.280 | 5.280 |
| 69 | 4.343 | 5.124 | 4.343 | 5.186 | 4.343 | 5.343 |
| 70 | 4.405 | 5.186 | 4.405 | 5.248 | 4.405 | 5.405 |
| 71 | 4.468 | 5.249 | 4.468 | 5.311 | 4.468 | 5.468 |
| 72 | 4.530 | 5.311 | 4.530 | 5.373 | 4.530 | 5.530 |
| 73 | | | 4.593 | 5.434 | 4.593 | 5.593 |
| 74 | | | 4.655 | 5.498 | 4.655 | 5.655 |
| 75 | | | 4.718 | 5.561 | 4.718 | 5.718 |
| 76 | | | 4.780 | 5.623 | 4.780 | 5.780 |
| 77 | | | 4.843 | 5.686 | 4.843 | 5.843 |
| 78 | | | 4.905 | 5.748 | 4.905 | 5.905 |
| 79 | | | | | 4.968 | 5.968 |
| 80 | | | | | 5.030 | 6.030 |
| 81 | | | | | 5.093 | 6.093 |
| 82 | | | | | 5.155 | 6.155 |
| 83 | | | | | 5.218 | 6.218 |
| 84 | | | | | 5.280 | b.280 |
| 85 | | | | | 5.343 | 6.343 |
| 86 | | | | | 5.405 | 6.405 |
| 87 | | | | | 5.468 | 6.468 |
| 88 | | | | | 5.530 | 6.530 |
| 89 | | | | | 5.593 | 6.593 |
| 90 | | | | | 5.655 | 6.655 |



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REQUIREMENTS:

Application criteria. The sleeve bolt shall be used to provide a precision net fit or controlled interference fit. The sleeve bolts shall be used in accordance with the limitations of MIL-B-85667/1. The sleeve bolts shall not be degreased and cetyl alcohol shall not be removed before installation.

Companion and accessory parts. Nuts in accordance with MIL-N-85729 shall be used with the sleeve bolts.

Oversize bolts. First and second oversize sleeve bolts may replace MIL-DTL-85667/12 sleeve bolts for first and second repair operations.

Installation. Installation shall be in accordance with MIL-HDBK-271/104.

Material. Pin material shall be titanium 6AL-4L in accordance with AMS 4967 or AMS 4928 (UNS R56400).

Sleeve material shall be corrosion resistant steel, AISI 304 (UNS S30400), in accordance with MIL-T-8504 (chemistry only) or ASTM A269 (chemistry only).

Heat treatment. Pin shall be heat treated in accordance with SAE-AMS-H-81200 to *95* Ksi Fsu minimum and 160 Ksi Ftu minimum.

Surface texture. Surface texture shall be 32 microinches Ra in accordance with ASME B46.1.

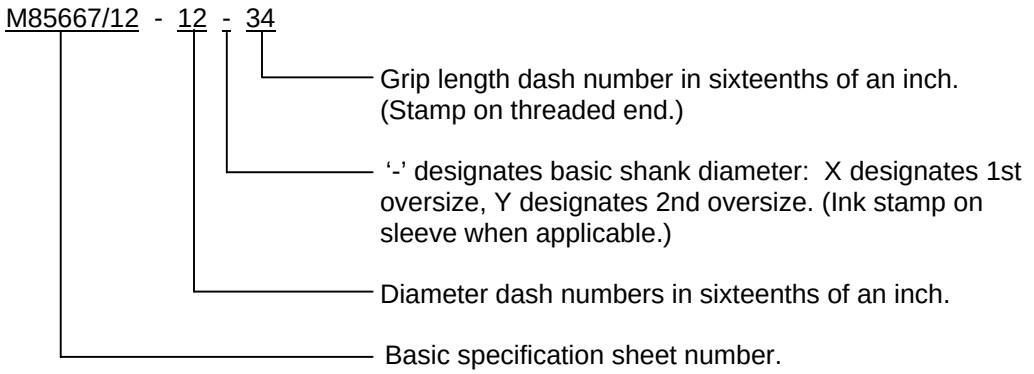
Coatings or surface treatment. Pin shall be dry-film lubricated in accordance with MIL-PRF-46010, Type I, .0003 / .0005 thick.

Sleeve - None.

Lubricants. .0001 inch overspray allowed on head and threads. After application of dry lubrication, the pin outside diameter and sleeve inside diameter shall be lubricated with cetyl alcohol in accordance with MIL-L-87132 or equivalent.

Unless otherwise specified, all dimensions are in inches. Tolerances: Decimals ± .010, angles +2°.

Example of military part number:






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In the event of a conflict between the text of this standard and the references cited herein, the text of this standard shall take precedence.

Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army – AV
Navy – AS
Air Force – 99

Preparing activity:
DLA – IS
(Project 5306-2722)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at www.dodssp.daps.mil.



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